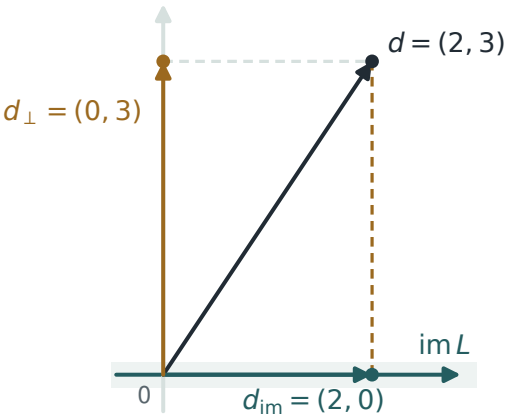
Correctable defect and cokernel residual

$L : \mathbb{R}^2 \to \mathbb{R}^2$, $L(a, b) = (a, 0)$. Defect $d = (2, 3)$.



A chosen scalar detector

$$\ell(x, y) = y$$

$$\ell L = 0$$

$$\ell(d) = 3 \neq 0$$

$$u = (-2, 0): \quad d + Lu = (0, 3) \neq 0.$$

The class $[d] \in \mathbb{R}^2/\operatorname{im} L$ is represented by $(0, 3)$.

Standard Euclidean splitting is used. The scalar 3 depends on the displayed quotient identification.